28 · The ear and sound: the fine print

The words of this track, as it says them. Bold brown is the examiner.

The ear and sound, the fine print. Her worksheets, her tables, and her orders.

The written tracks walk you through the ear and through waves. Her worksheets and matching tables ask for more of the small parts, and for things in a particular order. This track is those.

The hook: air, bone, fluid. The oval window pushes, the round window gives. Low potassium outside, high potassium inside, and glutamate fires the nerve.

The outer ear. The pinna is made of cartilage, with the ear lobe at the bottom. It picks up sound waves and channels them down the ear canal, which her worksheet calls the external auditory meatus. The ceruminous glands in the canal make cerumen, a yellow-brown wax. If it packs the canal, the person struggles to hear: conduction deafness.

The middle ear. The middle ear cavity sits in the temporal bone. The tympanic membrane, the eardrum, is a relatively large, tough membrane that turns vibrations in air into vibrations in bone. The ossicles act as levers, to amplify the vibrations, and the stapes is the smallest bone in the human body. The Eustachian tube connects the middle ear to the throat, and equalises the pressure on the eardrum.

Her true or false: in the outer ear, sound travels through air; in the middle ear, through bone; and in the inner ear, through fluid. True. The oval window is where vibrations in bone become vibrations in fluid. And the round window: the oval window and the round window vibrate in opposite directions. As the stapes pushes the oval window in, the round window bulges out, so the fluid has room to move. Her order of the membranes: the tympanic membrane, the oval window, the basilar membrane, then the round window.

The inner ear. The cochlea is a fluid-filled, coiled tube, lined with the sensory cells of the organ of Corti, which turn vibrations in fluid into electrical impulses. The semicircular canals are fluid-filled, and set at right angles to each other, so they pick up movement in every direction. Two nerves leave the inner ear: the vestibular nerve, for balance, and the cochlear, or auditory, nerve, for hearing, which carries the impulses as action potentials to the temporal lobes.

Her organ of Corti sheet. Perilymph fills the scala vestibuli and the scala tympani, and it's low in potassium. Endolymph fills the scala media, and it's high in potassium. When the hairs bend, potassium and calcium flow into the top of the hair cell, and it releases the neurotransmitter glutamate, which sends an impulse on to the auditory cortex. Her other wording: the stereocilia bend against the tectorial membrane, excitatory neurotransmitter is released at the synapse with the sensory neurons, and action potentials travel to the central nervous system. Mechanical in, electrical out.

Her orders. Her worksheet's seven steps: movement of the oval window, movement of the perilymph in the scala tympani, movement of the basilar membrane, movement of the endolymph in the scala media, bending of the sensory hairs as the gel-like tectorial membrane moves, impulses along the cochlear nerve, and the auditory cortex in the temporal lobe. Notice that her orders put the basilar membrane before the endolymph. In her arrange-the-steps questions, use her order.

Pitch along the cochlea. Low sounds move the longer, floppier fibres, at the apex of the cochlear duct, furthest from the oval window. High-pitched sounds move the shorter, stiffer hair cells, at the base. Her receptors table says the same: high pitch, hair cells at the base of the cochlear duct; low pitch, hair cells at the apex.

Her light-and-sound table, ten marks, in two quizzes. Sound waves are mechanical waves: vibrations of particles, so they can't travel through a vacuum. Light waves are electromagnetic waves, so they can. Photons are light particles, or packages of light. The amplitude of light gives its intensity, how bright it is, measured in lux. The amplitude of sound gives its loudness, measured in decibels. The frequency and wavelength of light give its colour: red light has the lowest frequency, and violet the highest. Of sound, its pitch: a higher frequency is a higher pitch. And all light waves travel at the same speed, about three hundred thousand kilometres per second.

Her wave table's definitions. Wavelength: the distance between the crests of the waves, measured in nanometres for light. Frequency: how often a wave passes a point each second. Amplitude: the maximum displacement, or height, of the wave from its resting point.

Two more refraction wordings. A wave bends because one part of it slows down, or speeds up, relative to another part. Refraction goes into the new medium, but changes direction; reflection returns to the same medium. Her example of refraction: an object in water looks as if it's in a different position, like a straw that looks bent. Her example of reflection: a mirror image.

And three of her sense words. Static and dynamic equilibrium both work by mechanoreception. Taste and smell are both chemoreception, and her key says taste is eighty per cent smell.

The traps. The round window moves the opposite way to the oval window. Endolymph is high in potassium, perilymph is low. Low pitch is at the apex, high pitch at the base. And loud sound is amplitude, measured in decibels; bright light is amplitude too, measured in lux.

Once more. The ear and sound, fine print.

The pinna is cartilage, with the ear lobe, and channels sound down the external auditory meatus, where ceruminous glands make wax that can cause conduction deafness. The eardrum turns air vibrations into bone vibrations, and the ossicles act as levers; the stapes is the smallest bone in the body. Air in the outer ear, bone in the middle, fluid in the inner. The oval window turns bone vibrations into fluid vibrations, and the round window moves the opposite way. The semicircular canals sit at right angles. The vestibular nerve is balance; the cochlear nerve is hearing.

Perilymph, in the scala vestibuli and tympani, is low in potassium. Endolymph, in the scala media, is high. Potassium and calcium flow into the hair cell, and it releases glutamate. Low pitch is at the apex, high pitch at the base. Sound is mechanical and can't cross a vacuum; light is electromagnetic, made of photons. Light's amplitude is its intensity, in lux; sound's is loudness, in decibels. Red light has the lowest frequency, violet the highest. Refraction bends a wave as part of it slows, like an object in water. Taste and smell are chemoreception, balance is mechanoreception, and taste is eighty per cent smell.